Array

Imp pgms

8) putting zero's at end

i/p → 1 0 2 0 3 0

o/p → 1 2 3 0 0 0

a[10] → b[10]

1. Size of array
2. Scan the ele or use rand
3.

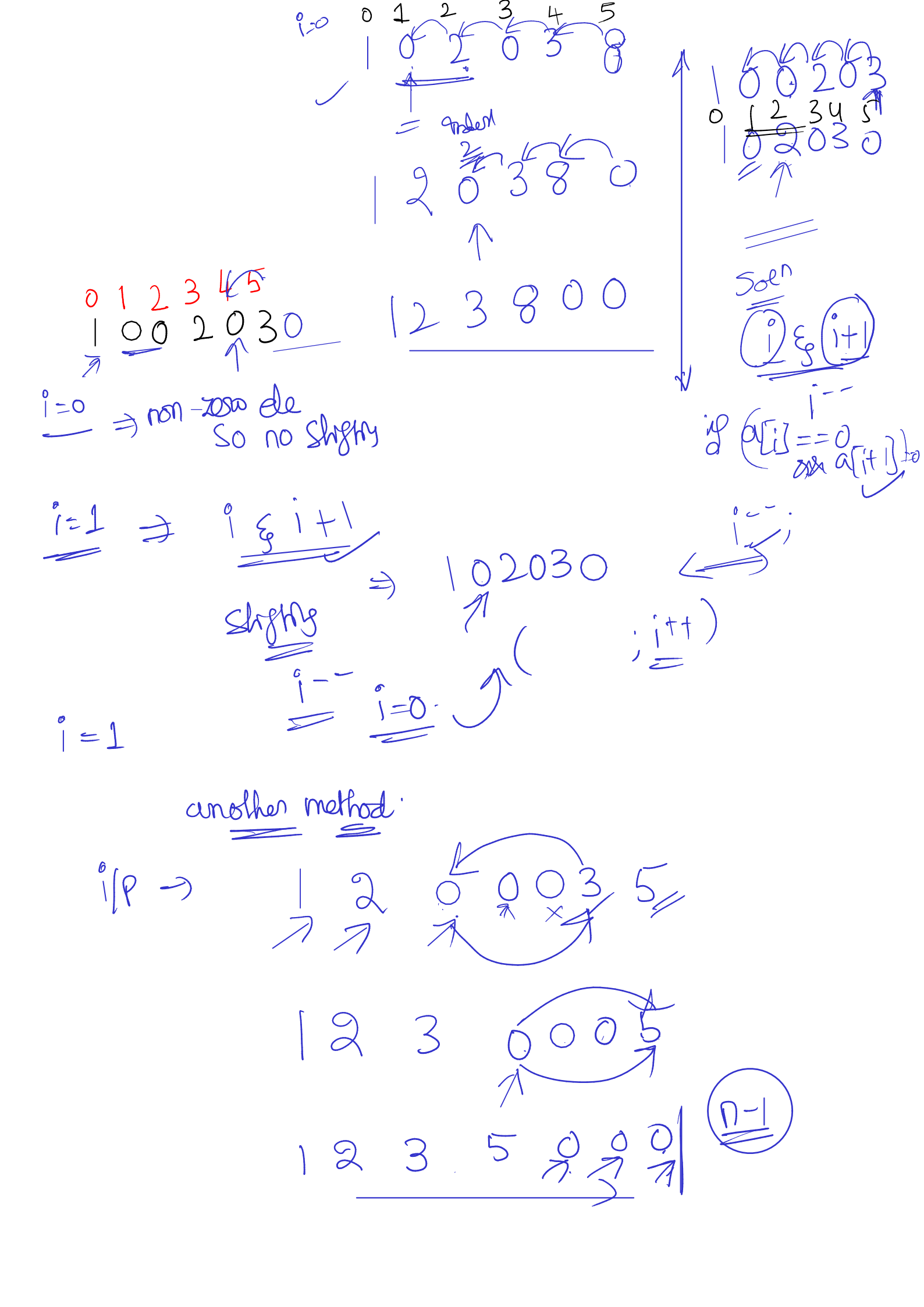
Logic : run a for loop
condn is if a[i]!=0
move the element into array b

```
int i, j=0
for(i=0; i<n; i++)
{
    if(a[i]!=0)
    {
        b[j] = a[i];
        j++;
    }
}
```

b[10]

{1, 2, 3, ... }

```
for(i=0; i<n; i++)
printf("%d", b[i]);
```


i=0
0 1 2 3 4 5
1 0 2 0 3 0
1 2 0 3 8 0
1 2 3 8 0 0
0 1 2 3 4 5
1 0 0 2 0 3 0
i=0 ⇒ non-zero de
So no shifting
i=1 ⇒ i & i+1
⇒ 1 0 2 0 3 0
shifting
i--
i=0
i=1
(; i++)
another method:
i/p →
1 2 0 0 3 5
1 2 3 0 0 0 5
1 2 3 5 0 0 0
n-1
Soln
i & i+1
i--
if a[i] == 0
a[i+1]

9) Finding max sum of Subarray

i/p → {1 2 3 4 5}

⇒

1 8 3 8 3

1+5=6

5+3=8

3+8=11

Subarray → 2

1+2=3

2+3=5

3+4=7

4+5=9

o/p → 9

Subarray → 3

1+2+3=6

2+3+4=9

3+4+5=12

o/p: 12

i/p: Size of main array → 5

ele - scan → 1 2 3 4 5

- read

Size of Subarray →

① to access every ele present in the array for loop & index i

② 2nd loop → iterate Subarray Size

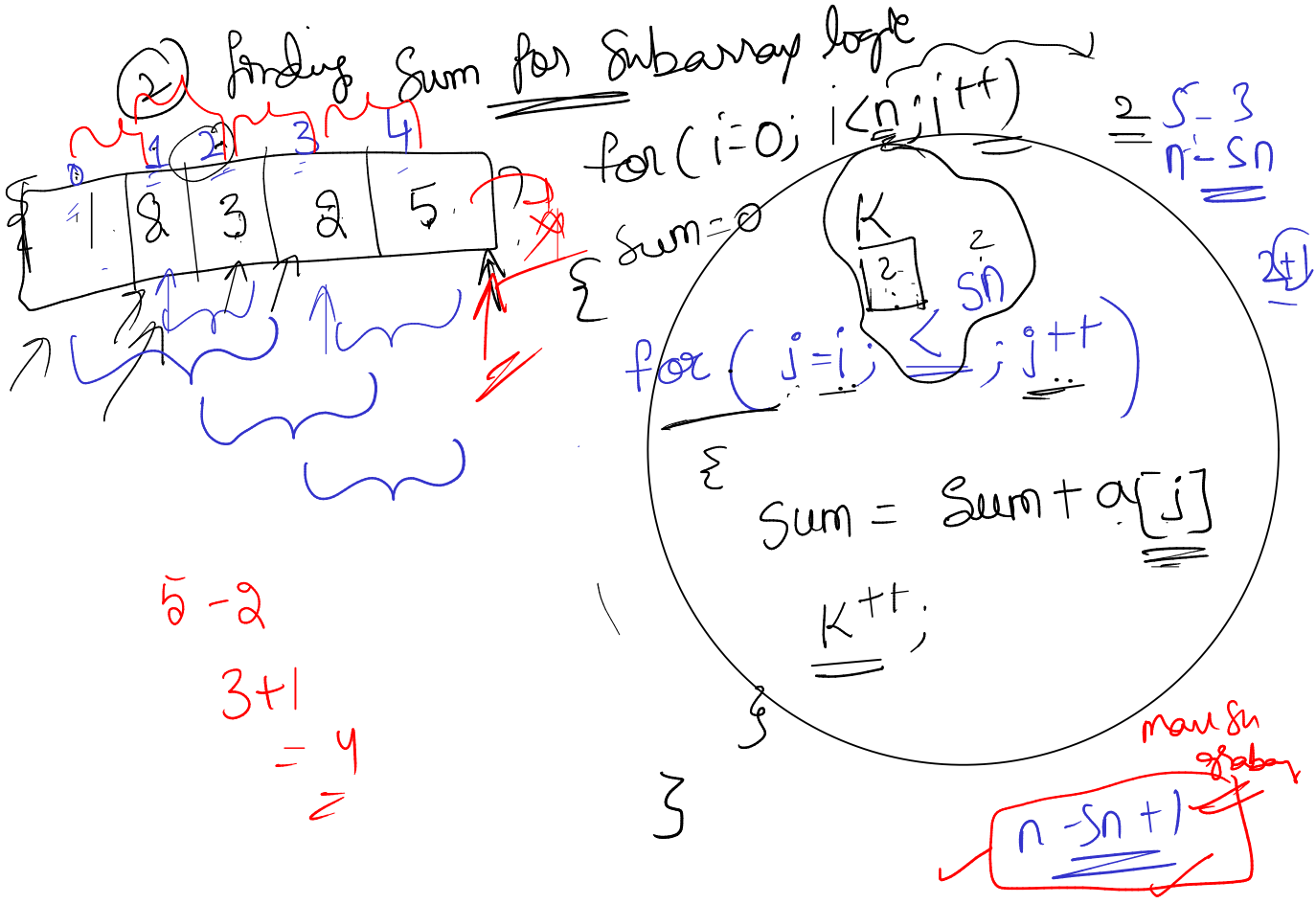
sum = 0

in loop itself sum the elements (count should be Subarray size)

(3) logic to check the present sum with previous sum & update according to the result

after coming out of the outer loop

(4) print the sum.

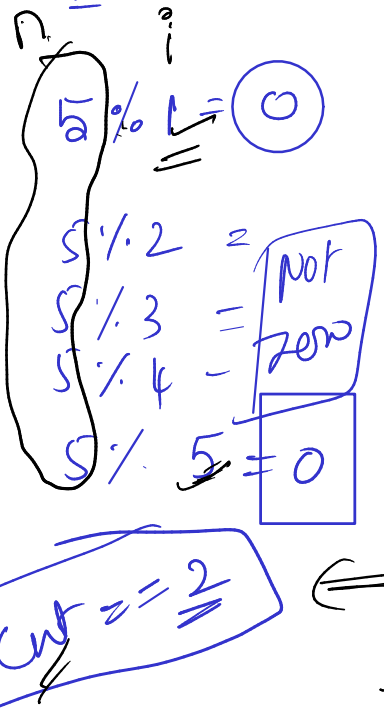


(8) prime & not-prime sorted

i/p: 1 2 3 4 5 6 7 8

2 3 | 1 4 5 6 8

5 → prime



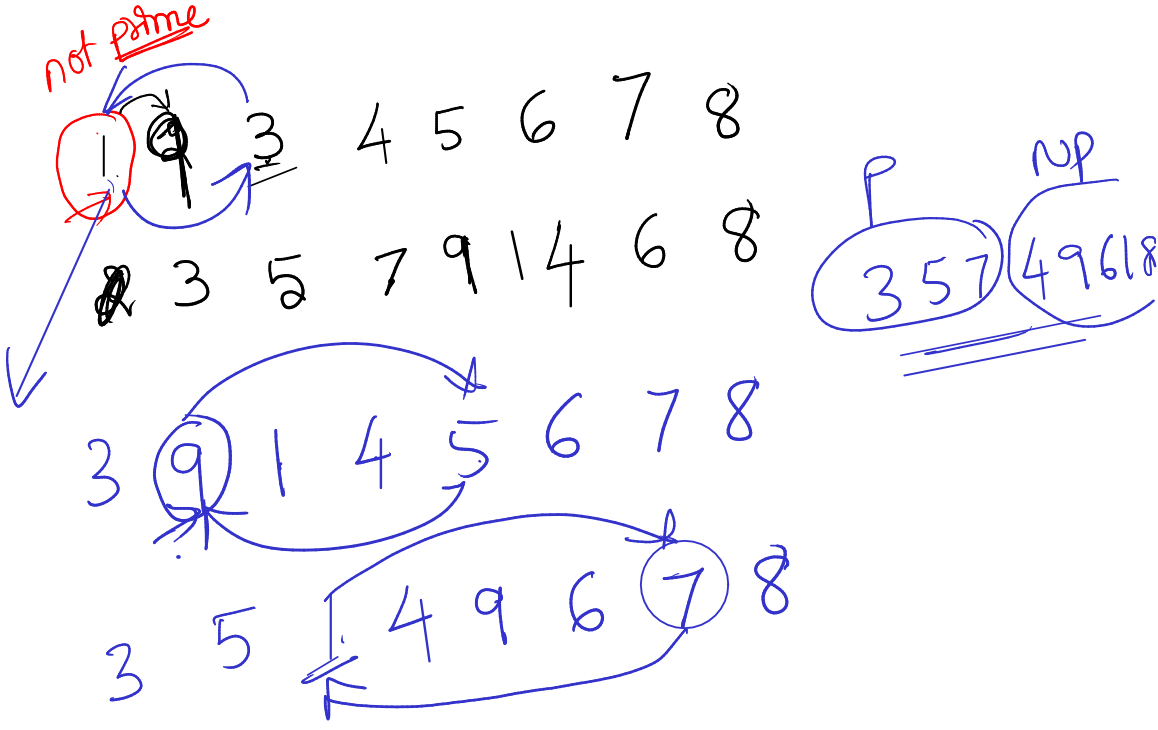
cnt == 2

6 → not prime

6 % 1 → 0
6 % 2 → 0
6 % 3 → 0
6 % 4
6 % 5
6 % 6 → 0

```
n = a[k]
for(i=1; i<=n; i++)
{
  if(a[k] % i == 0)
    cnt++
}
if (cnt == 2)
  printf("prime");
```

o/p



Q) Wap to find highest element out of 3 diff arrays

I/P a[5] = { 1, 8, 5, 6, 2 }

b[5] = { 23, 82, 56, 32 }

c[5] = { 0, 1, 8, 87, 52 }

O/P ⇒ 87